Introduction

Priority Queue : using other implementations Data Structures and its Applications (UE25CS52A)

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Quiz : Priority Queue : using other implementations

1. Which statement best describes enque in a priority queue implemented using an unordered array?

a) Can add in the beginning; would require a few operations independent of the number of elements in the array
b) Can add at the end; would require a few operations independent of the number of elements in the array
c) Should insert in the middle
d) Should depend on the priority

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Quiz : Priority Queue : using other implementations

2. Which statement best describes enque in a priority queue implemented using an ordered array?

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Quiz : Priority Queue : using other implementations

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Quiz : Priority Queue : using other implementations

3. Which statement best describes enque and deque in a priority queue implemented using an ordered list?

a) Both require operations proportional to the number of elements in the priority queue
b) Both require a few operations independent of the number of elements in the priority queue
c) Enque would require more operations than deque
d) Deque would require more operations than enque

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Quiz : Priority Queue : using other implementations

3. Which statement best describes enque and deque in a priority queue implemented using an ordered list?

a) Both require operations proportional to the number of elements in the priority queue
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c) Enque would require more operations than deque
d) Deque would require more operations than enque

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Quiz : Priority Queue : using other implementations

4. Deinitializing a priority queue requires

a) A Few operations in an array implementation
b) A few operations in a list implementation
c) A number of operations proportional to the number of elements in an array implementation
d) Freeing of resources in array implementation

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Quiz : Priority Queue : using other implementations

4. Deinitializing a priority queue requires

a) A Few operations in an array implementation
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c) A number of operations proportional to the number of elements in an array implementation
d) Freeing of resources in array implementation

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Quiz : Priority Queue : using other implementations

5. Deque in an unordered array implementation of priority queue requires

a) Shifting the elements from right to left
b) Shifting the elements from left to right
c) Just a single assignment – no shifting
d) None of these

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Quiz : Priority Queue : using other implementations

5. Deque in our unordered array implementation of priority queue requires

a) Shifting the elements from right to left
b) Shifting the elements from left to right
c) Just a single assignment – no shifting
d) None of these

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Priority Queue: concept

- A priority queue is an abstract data type in which each element is associated with a priority.
- A priority queue removes the component with the highest priority first.
- A priority is a number associated with each element
- Higher the numeric value, could mean higher the priority.
- Or lower the numeric value, could mean higher priority

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Priority Queue: some examples

- Patient treatment
- Scheduling of processes

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Priority Queue: interface

initialize() deinitialize()	Create an empty queue Make the queue empty
enqueue(x) I	Insert x We may add in number of ways. This in turn would affect dequeue
dequeue()	Remove the element with the highest priority. This in turn depends how enqueue operation was done
isEmpty()	Check whether the queue is empty
isFull()	Check whether the queue is full

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Priority Queue: Possible Implementations

- Unodered list
- Ordered list : ordered based on priority
- Unordered array
- Ordered array : ordered based on priority
- Heap tree

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Priority Queue: ordered list

- Implemented using queue implementation
- Could also implement using a list

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Priority Queue: ordered list implementation

- Initialize : make the front and the rear NULL
- Deinitialize : deallocate the nodes and then make the front and the rear NULL
- Is_empty : check front is NULL
- Is_full : check whether memory allocation fails

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Priority Queue: ordered list implementation

- Enque :
  - Make a node
  - Populate
  - Find the position to insert
  - Change front if required
  - Numer of operations depends on the number of elements in the priority queue
- Deque :
  - Copy the job to return
  - Change the front
  - Release the node
  - Suitably adjust front and then rear if required

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Priority Queue: unordered array implementation

- Initialize : make the number of elements in the array 0
- Deinitialize : same as initialize!
- Is_empty : check if the number of elements is 0
- Is_full : check if the number of elements equal to the size of the array

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Priority Queue: unordered array implementation

- Enque :
  - Check if full
  - Add at the end
  - Increase the size
  - Only a few operations
- Deque :
  - Traverse from the beginning
  - Find the leftmost position of the node with the highest priority
  - Copy the job
  - Shift right to left from this position till the end
  - Decrease the size
  - Return the job

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Priority Queue: ordered array implementation

- Initialize : make the number of elements in the array 0
- Deinitialize : same as initialize!
- Is_empty : check if the number of elements is 0
- Is_full : check if the number of elements equal to the size of the array

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Priority Queue: ordered array implementation

- Enque :
  - Check if full
  - Traverse right to left; shift; find the position to insert
  - insert
  - Increase the size
  - Number of operations proportional to the number of elements
- Deque :
  - Check if empty
  - Find the rightmost element
  - Decrement the size
  - Return the job of the rightmost element before changing size

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Priority Queue: Applications of priority queue :

- CPU Scheduling of processes
- Hospital Emergency Services
- Printer Scheduling
- Network Packet Scheduling
- Dijkstra's Shortest Path Algorithm
- Prim's Minimum Spanning Tree Algorithm
- Event-Driven Simulation
- Huffman Coding

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THANK YOU

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